

Algebra 1
Unit 13
Lesson 11 Practice

Name Answer Key
Date _____
Period _____

The table shows the number of tons of blueberries that were imported to the US from 2005 to 2013.

Year	Blueberries Imported (tons)
2005	26,319
2006	32,602
2007	35,125
2008	52,111
2009	60,450
2010	76,743
2011	87,336
2012	96,367
2013	103,502

1. Determine the average rate of change from 2010 to 2013.

$$\begin{array}{l} (2010, 76743) \\ (2013, 103502) \end{array} \quad \frac{y_2 - y_1}{x_2 - x_1} = \frac{103,502 - 76,743}{2013 - 2010} = \frac{26,759}{3} \approx 8,919.67$$

2. Interpret the average rate of change for the interval 2010 to 2013 in terms of the context.

Between the years 2010 to 2013 the amount of blueberries imported increased 8,919.67 tons per year, on average.

3. Determine the average rate of change from 2005 to 2009.

$$\begin{array}{l} (2005, 26319) \\ (2009, 60450) \end{array} \quad \frac{y_2 - y_1}{x_2 - x_1} = \frac{60,450 - 26,319}{2009 - 2005} = \frac{34,131}{4} = 8,532.75$$

4. Interpret the average rate of change for the interval from 2005 to 2009 in terms of the context.

Between the years of 2005 and 2009, an average of 8,532.75 more tons of blueberries were imported each year.

5. How does the average rate of change for the interval from 2005 to 2009 compare to the average rate of change for the interval 2010 to 2013?

The average rate of change from 2005-2009 is fewer by 386.92 compared to 2010-2013.

The function $M(x) = 5x^2 - 160x + 6179$ models the number of miles, in millions, U.S. passengers traveled on a train for the period of 1970 to 2000, where x is the number of years since 1970.

6. What is the average rate of change, in miles traveled by U.S. passengers per year, for the period 1980 to 1985?

$$\begin{array}{rclcl}
 & 5(10)^2 - 160(10) + 6,179 & 5(15)^2 - 160(15) + 6,179 & \frac{y_2 - y_1}{x_2 - x_1} & \\
 1980 - 1970 = 10 & 500 - 1,600 + 6,179 & 1,125 - 2,400 + 6,179 & & \\
 1985 - 1970 = 15 & -1,100 + 6,179 & -1,275 + 6,179 & \frac{4,904 - 5,079}{15 - 10} & \\
 & = 5,079 & = 4,904 & & \\
 & (10, 5079) & (15, 4904) & \frac{-175}{5} & \boxed{= -35}
 \end{array}$$

7. Interpret the average rate of change for the interval 1980 to 1985 in terms of the context.

From 1980 to 1985 the average rate of change in regards to passengers traveling by train was -35 million miles. So, on average there was a decrease in train travel during this time period.

8. What is the average rate of change, in miles traveled by U.S. passengers per year, for the period 1985 to 1990?

$$\begin{array}{rclcl}
 & 5(20^2) - 160(20) + 6,179 & \frac{y_2 - y_1}{x_2 - x_1} & & \\
 1985: (15, 4904) & 2,000 - 3,200 + 6,179 & & & \\
 1990: 1990 - 1970 = 20 & -1,200 + 6,179 & \frac{4,979 - 4,904}{20 - 15} & & \\
 & = 4,979 & & & \\
 & (20, 4979) & \frac{75}{5} & \boxed{= 15} &
 \end{array}$$

9. Interpret the average rate of change for the interval 1985 to 1990 in terms of the context.

From 1985 to 1990 the average rate of change in regards to passengers traveling by train was 15 million miles. So, on average there was an increase in train travel during this time period.

10. The graph of a function is shown. Points $(2, 2.25)$ and $(5, 7.594)$ are on the function. Calculate the average rate of change for the line that passes through the points $(2, 2.25)$ and $(5, 7.594)$.

$$\frac{y_2 - y_1}{x_2 - x_1} = \frac{7.594 - 2.25}{5 - 2} = \frac{5.344}{3} \approx 1.781\bar{3}$$

